THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sum of the first 8 odd natural numbers is: 1

(a) 63 **(b)** 72 **(c)** 64 **(d)** 16

2. The smallest number by which 54 must be multiplied to get a perfect cube is: 1

(a) 4 **(b)** 2 **(c)** 6 **(d)** 3

3. If $2^x = 64$ then x is: 1

(a) 7 **(b)** 8 **(c)** 5 **(d)** 6

4. The Roman numeral CDXLIV stands for: 1

(a) 445 **(b)** 434 **(c)** 454 **(d)** 444

5. Which quadrilateral has two pairs of equal adjacent sides? 1

(a) kite **(b)** parallelogram **(c)** rectangle **(d)** trapezium

6. The diagonal of a rectangle with sides 5 cm and 12 cm is: 1

(a) 14 cm **(b)** 13 cm **(c)** 12 cm **(d)** 17 cm

7. The HCF of 24 and 36 is: 1

(a) 12 **(b)** 6 **(c)** 24 **(d)** 72

8. The value of $105^2 - 95^2$ is: 1

(a) 40000 **(b)** 100 **(c)** 2000 **(d)** 20050

9. If 3 : 5 = 12 : x, then x equals: 1

(a) 21 **(b)** 20 **(c)** 19 **(d)** 23

10. The ratio 36 : 90 in its simplest form is: 1

(a) 2 : 6 **(b)** 2 : 5 **(c)** 5 : 2 **(d)** 3 : 5

11. The compound interest on a sum for 2 years is greater than the simple interest because: 1

(a) time is longer
(b) interest is also earned on earlier interest
(c) rate is higher
(d) principal increases at start

12. The simple interest on ₹ 5000 at 8% per annum for 2 year(s) is: 1

(a) ₹ 400 **(b)** ₹ 800 **(c)** ₹ 1600 **(d)** ₹ 5800

13. Is a triangle with sides 6, 8 and 10 a right triangle? 1

(a) Cannot say **(b)** No
(c) Yes **(d)** Only if it is isosceles

14. The legs of a right triangle are 6 cm and 8 cm. Its hypotenuse is: 1

(a) 11 cm **(b)** 10 cm **(c)** 9 cm **(d)** 14 cm

15. A scale drawing has scale 1 : 50. A 2 m wall is drawn with length: 1

(a) 40 cm **(b)** 4 cm **(c)** 10 cm **(d)** 2 cm

16. If a cube is cut by a plane parallel to a face, the cross section is a: 1

(a) circle **(b)** square **(c)** triangle **(d)** hexagon

17. If every value in a data set is increased by 5, the mean: 1

(a) increases by 5 **(b)** doubles **(c)** increases by 25 **(d)** stays the same

18. The sum of any two consecutive integers n and $n+1$ is: 1

(a) prime **(b)** a multiple of 4 **(c)** even **(d)** odd

19. The area of a rhombus with side 5 cm and one diagonal 6 cm is: 1

(a) 48 cm² **(b)** 12 cm² **(c)** 30 cm² **(d)** 24 cm²

20. Which of the following can be the same for two shapes with different areas? 1

(a) angles **(b)** length only **(c)** colour **(d)** perimeter

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. What is the speed of light, 300000000 m/s, in scientific notation? 2

22. State the place values of digits in 734_8. 2

23. State whether 1001 is divisible by 7, 11 and 13. 2

24. What is direct proportion? Give an example. 2

25. What does Baudhayana's rectangle-diagonal rule say? 2

26. A sum of ₹ 900 is shared in the ratio 4 : 5 : 6. Find the smallest share. 2

27. Explain 'mean as a balancing point'. 2

28. Find the side of a square whose area is 225 m². 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Find the least square number divisible by each of 6, 8 and 12. 3

30. Show that diagonals of a rectangle are equal using Pythagoras. 3

31. Multiply $(3x+2)(2x-5)$. 3

32. Find the rate of simple interest if ₹ 8000 earns ₹ 1920 interest in 3 years. 3

33. Match each solid with its top view: cube, cylinder, cone, square pyramid. 3

34. A number pyramid has the top 25 and the bottom row a, 6, c with $a = 7$. Find c. 3

35. A flagpole is supported by a 15 m wire fixed 9 m from its foot. Find the height at which the wire is attached. 3

OR

Find the perimeter of a rhombus whose diagonals are 16 cm and 30 cm. 3

36. A square of side 14 cm has a smaller square of side 6 cm removed from a corner. Find the remaining area. 3

OR

A triangular field has base 25 m and height 14 m. Find its area and the cost of levelling at ₹ 12 per m². 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. **Case study:** The population of India is about 140 crore and area is about 32.9 lakh km². 5

(i) Write both in standard form.

(ii) Find the approximate population density.

(iii) Compare with a city having density 11000 per km².

OR

(i) Write 4.7×10^{-4} in decimal form. 5

(ii) Write 0.00000092 in standard form.

(iii) Add $3 \times 10^4 + 5 \times 10^3$ and express in standard form.

38. **Case study:** Meera deposits ₹ 25000 in a bank at 8% compound interest annually. 5

(i) Find the amount after 2 years.

(ii) The interest earned.

(iii) If simple interest were paid instead, what would be the difference?

OR

(i) A trader marks goods 40% above cost and allows 25% discount. Find the profit percent. 5

(ii) If CP is ₹ 500, find the SP.

39. (i) How is a dot plot made? 5

(ii) Construct one for marks 6, 7, 7, 8, 8, 8, 9, 9, 10 and read the mean and median.

OR

Case study: A line graph shows the daily temperature: Mon 28, Tue 30, Wed 33, Thu 31, Fri 29, Sat 27, Sun 30. 5

(i) Find the mean.

(ii) On which day is the temperature highest?

(iii) Find the range and describe the trend.

40. (i) Show that a median divides a triangle into two equal areas. 5

(ii) The area of triangle ABC is 80 cm²; D is the midpoint of BC. Find the area of triangle ABD.

OR

Case study: A rectangular hall 12 m × 8 m has a carpet in the middle leaving a 1 m border. 5

(i) Area of the carpet.

(ii) Area of the border.

(iii) Cost of polishing the border at ₹ 55/m².

END OF QUESTION PAPER Total: 80 Marks